

Wireless Security and Configuration — Guided Notes

Unit 2.2.2 | CompTIA Network+ N10-009 Obj. 2.2 | N10-008 2.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

WEP is broken because its _____ -bit initialization vector space is too small, causing IVs to _____. This allows an attacker to crack the key in under _____. WEP should _____ be deployed.

Question 2

WPA used _____ as its encryption protocol, which wrapped around the same _____ cipher as WEP. WPA2 replaced this with _____, which uses the _____ block cipher with 128-bit keys.

Question 3

WPA3 replaces the PSK handshake with _____ (SAE), which makes offline brute-force attacks _____ because the attacker must _____ for each guess. WPA3 also provides _____, meaning captured traffic cannot be decrypted even if the password is later compromised.

Question 4

WPA2/WPA3 Personal uses a _____ shared by all users. Enterprise uses _____ where each user authenticates with _____ credentials. The AP passes authentication to a _____ server which checks against a directory.

Question 5

Enterprise mode advantages over Personal include: (1) _____, (2) _____, (3) _____, and (4) _____.

Question 6

On 2.4 GHz, the only three non-overlapping channels are _____, _____, and _____. Using channel 3 causes _____ interference which is _____ than using the same channel as a neighbor

because devices can't decode the signal cleanly but still wait for it.

Question 7

A rogue AP is _____ connected to your network without authorization. An evil twin is _____ that mimics a legitimate network. Evil twin attacks are more dangerous against _____ mode networks because clients have no way to verify the AP's _____.

Question 8 — Real World Application

REAL WORLD: A teacher reports that their laptop keeps connecting to a network called 'STAFF' but then has no internet access and gets strange certificate warnings. Other staff are unaffected. What attack is likely occurring and what steps do you take?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-2-2-wireless-security-and-configuration/2-2-2-wireless-security-and-configuration-answer-key.pdf
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